

GAYATHRI KARAGODA PATHIRANAGE

Feltham, UK : madushanikpg557@gmail.com : 07952325029

LinkedIn: [linkedin.com/in/gayathri-pathiranage/](https://www.linkedin.com/in/gayathri-pathiranage/)

Portfolio: https://gayathri324.github.io/Portfolio_Gayathri_Pathiranage/

PROFILE

Data Analyst with an MSc in Statistics and Data Analytics, skilled in SQL, Python, and data visualization. Experienced in transforming large datasets into actionable insights through statistical analysis, predictive modelling, and dashboard development. Strong ability to communicate insights and support data-driven decision-making.

TECHNICAL SKILLS

Programming & Databases: Python (Pandas, NumPy, Matplotlib), SQL (MySQL, SQL Server), R

Data Visualization: Tableau, Power BI, Excel (Pivot Tables, VLOOKUP)

Statistical & Analytical Skills: Regression (Linear, Logistic), Bayesian Inference, Predictive Modelling, Time Series (ARIMA)

Tools & Techniques: EDA, Data Cleaning & Preprocessing, Dashboard Development, Data Storytelling

PROJECTS

UK Rail Performance & Delay Analysis Dashboard

- Analyzed 30,000+ train journey records using SQL and Python
- Built an interactive Tableau dashboard for travel demand, delays, and revenue
- Identified peak delay periods and key causes such as weather and signal failures
- **Impact:** Provided actionable insights to improve scheduling and operational efficiency

England & Wales Property Price Analysis

- Performed SQL-based analysis on property transaction data
- Developed Tableau dashboard for regional price trends and comparisons
- Analyzed property types, tenure, and long-term price growth
- **Impact:** Revealed regional disparities and key market trends

Apple Stock Price Analysis & Forecasting

- Built predictive models using ARIMA and regression in Python
- Compared model performance using MSE
- Visualized forecasts using Power BI
- **Impact:** Improved short-term forecasting accuracy using lag-based models

Energy Consumption Prediction (Bayesian Logistic Regression)

- Developed Bayesian model using MCMC in R
 - Predicted probability of high energy consumption
 - Evaluated model performance using classification metrics
 - **Impact:** Enabled prediction under uncertainty using probabilistic modelling
-

EDUCATION

MSc Statistics with Data Analytics - Brunel University London, UK - 2023

BSc Mathematics, Mathematical Modelling & Statistics - Wayamba University of Sri Lanka - 2017